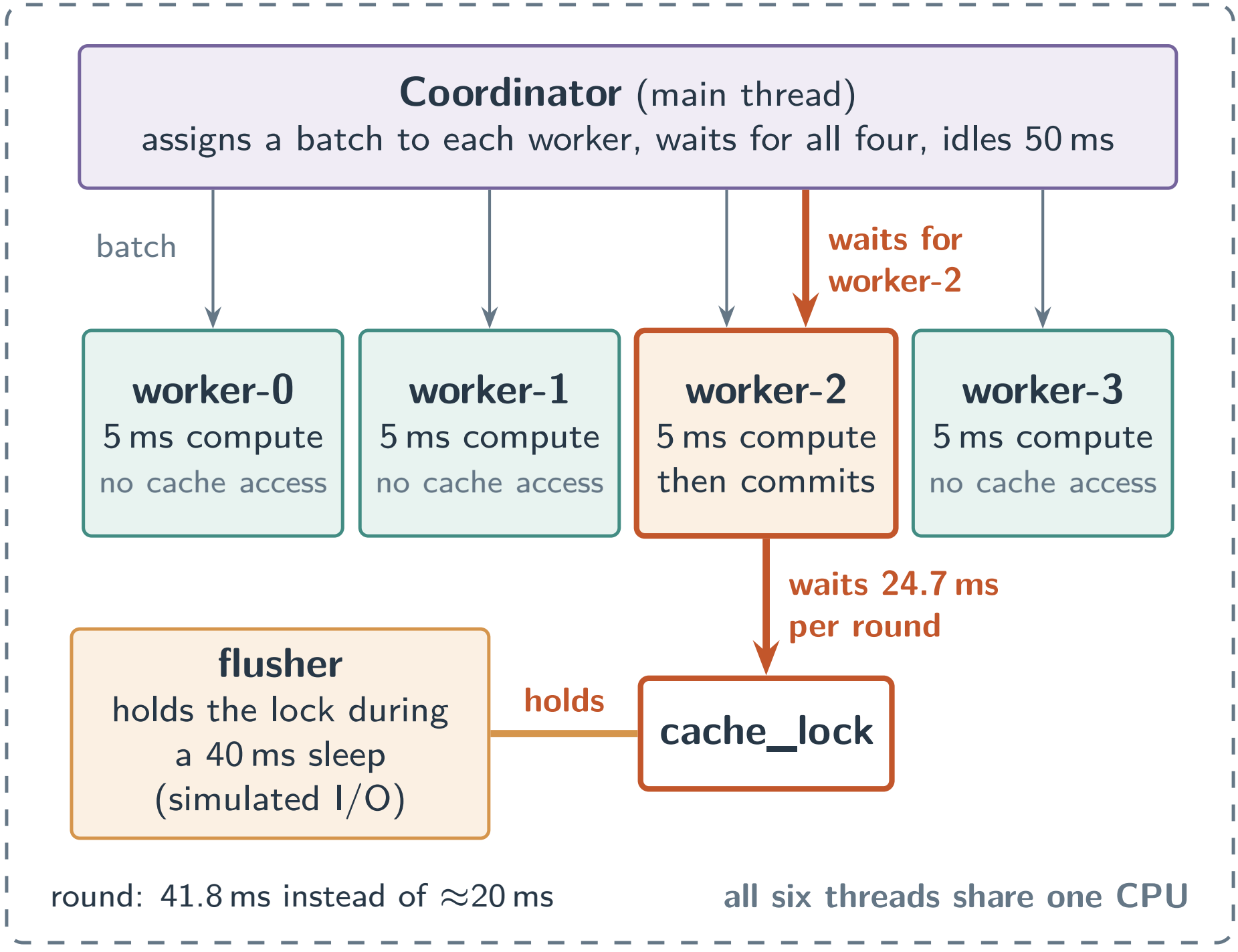
Coordinator (main thread)
assigns a batch to each worker, waits for all four, idles 50 ms
batch
waits for worker-2
worker-0
5 ms compute
no cache access
worker-1
5 ms compute
no cache access
worker-2
5 ms compute
then commits
worker-3
5 ms compute
no cache access
waits 24.7 ms per round
flusher
holds the lock during a 40 ms sleep (simulated I/O)
holds
cache_lock
round: 41.8 ms instead of ≈20 ms
all six threads share one CPU